

The Caterpillar Factorization of Crowns

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Abstract

In this paper we obtain the necessary and sufficient condition for balanced-caterpillar factorization of crowns and that for directed caterpillar factorization of symmetric crowns.

1 Introduction and preliminaries

Suppose that G and H are graphs. A G -factor of H is a spanning subgraph of H which is the union of vertex disjoint subgraphs each isomorphic to G . A G -factorization of H is a set of G -factors of H which partition the edges of H . For digraphs G and H , a G -factor and a G -factorization of H are similarly defined.

For a graph H , the *symmetric digraph* H^* is the digraph obtained from H by replacing each edge of H by two arcs with opposite directions.

A star factorization is a G -factorization where G is a star. The necessary and sufficient conditions for the star factorization of K_n was obtained by Yu [14], and those of $K_{n,n}$ was obtained by Martin [6]. Wang [13], Martin [5, 7], Du [2, 3], and Du and Wang [4] investigated the star factorization of $K_{m,n}$, respectively. Ushio obtained the necessary and sufficient conditions for the

directed star factorization of $\lambda K_{m,n}^*$ [8, 12], and investigated that of $K_{l,m,n}^*$ [9, 10, 11]. Bryant et al. [1] gave the necessary and sufficient conditions for the star factorizations of some graph products.

A *caterpillar* is a tree of order at least three which contains a path such that each vertex not on the path is adjacent to a vertex on the path. For the convenience of discussions, a caterpillar can be defined alternatively as follows. For a positive integer d , let S_d denote a star with d edges. For positive integers $d_1, d_2, \dots, d_v$ ($v \geq 1$), $S(d_1, d_2, \dots, d_v)$ denotes the graph obtained from the stars $S_{d_1}, S_{d_2}, \dots, S_{d_v}$ by identifying an end-vertex of S_{d_i} with the center of $S_{d_{i+1}}$ for $i = 1, 2, \dots, v-1$. As an illustration, $S(5, 2, 1, 3)$ is exhibited in Fig. 1. It is easy to see that the graph $S(d_1, d_2, \dots, d_v)$ has $1 + d_1 + d_2 + \dots + d_v$ vertices. A caterpillar can be defined as a graph $S(d_1, d_2, \dots, d_v)$ with order $1 + d_1 + d_2 + \dots + d_v \geq 3$.

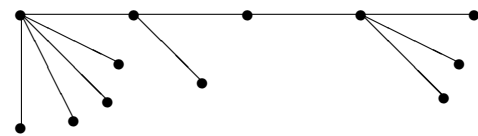


Fig. 1. $S(5, 2, 1, 3)$

For positive integers $k \leq n$, the *crown* $C_{n,k}$ is the graph with vertex set $\{a_0, a_1, \dots, a_{n-1}, b_0, b_1, \dots, b_{n-1}\}$ and edge set $\{a_i b_j : 0 \leq i \leq n-1, j = i+1, i+2, \dots, i+k \pmod{n}\}$.

In this paper, we study the caterpillar factor-

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ization of $C_{n,k}$ and the directed caterpillar factorization of $C_{n,k}^*$. We begin with some lemmas.

Lemma 1.1 *Suppose that G is a graph (digraph, respectively) of order n_1 , size e_1 and H is a graph (digraph, respectively) of order n_2 , size e_2 . We have the following*

- (1) *If H has a G -factor, then $n_2 \equiv 0 \pmod{n_1}$.*
- (2) *If H has a G -factorization, then $n_1 e_2 \equiv 0 \pmod{n_2 e_1}$.*

Proof. Let t be the number of components in a G -factor of H . Then $n_2 = tn_1$. Thus

- (1) $n_2 \equiv 0 \pmod{n_1}$.
- (2) A G -factor of H has $te_1 = \frac{n_2}{n_1}e_1$ edges. Let r be the number of G -factors in the factorization. Then $e_2 = r(\frac{n_2}{n_1}e_1)$, which implies $r = \frac{n_1 e_2}{n_2 e_1}$. Thus $n_1 e_2 \equiv 0 \pmod{n_2 e_1}$. $\square$

A connected bipartite graph is *balanced* if it has a bipartition (C, D) with $|C| = |D|$. It is easy to see that $S(d_1, d_2, \dots, d_v)$ is balanced if and only if $d_1 + d_3 + d_5 + \dots = 1 + d_2 + d_4 + \dots$.

Lemma 1.2 *Suppose that G is an unbalanced connected bipartite graph of order n_1 and H is a balanced connected bipartite graph of order $2n_2$. If H has a G -factor, then $n_2 \equiv 0 \pmod{n_1}$.*

Proof. Assume that (A, B) is a bipartition of G . Let $a = |A|$, $b = |B|$. Then $a + b = n_1$, $a \neq b$. Assume that (C, D) is a bipartition of H with $|C| = |D| = n_2$. Suppose that in a G -factor of H , there are l copies of G with A -part in C , and m copies of G with B -part in C . Then $la + mb = n_2$ and $lb + ma = n_2$. Thus $bn_2 - an_2 = m(b^2 - a^2)$. Since $a \neq b$, we have $n_2 = m(b + a)$. Thus $n_2 \equiv 0 \pmod{n_1}$. $\square$

Lemma 1.3 *Suppose that n, p, q are positive integers such that $pq \leq n$. Let G be a graph (digraph, respectively). If the crown $C_{n,p}$ (the symmetric crown $C_{n,p}^*$, respectively) has a G -factorization, then $C_{n,pq}$ ($C_{n,pq}^*$, respectively) has a G -factorization.*

Proof. It is easy to see that $C_{n,pq}$ ($C_{n,pq}^*$, respectively) can be decomposed into q spanning subgraphs (subdigraphs, respectively) of which each is isomorphic to $C_{n,p}$ ($C_{n,p}^*$, respectively). Since, by assumption, $C_{n,p}$ ($C_{n,p}^*$, respectively) has a G -factorization, so does $C_{n,pq}$ ($C_{n,pq}^*$, respectively). $\square$

2 Caterpillar factorization of crowns

In this section we investigate the caterpillar factorization of crowns. In the following, a_i and b_j are the vertices of $C_{n,k}$ as given in the definition, and the subscripts of a_i and b_j are always taken modulo n . Each edge $a_i b_{i+t}$ ($1 \leq t \leq k$) in $C_{n,k}$ is assigned the label t .

Lemma 2.1 *Let $G = S(d_1, d_2, \dots, d_v)$ be a caterpillar of order t . Let n be a positive integer with $n \geq t - 1$. Then there exists a subgraph G' of the crown $C_{n,t-1}$ such that*

- (i) $G' \cong G$,
- (ii) *the vertex set of $G' = \{a_i : i = 0, -1, -2, \dots, -d_2 - d_4 - d_6 - \dots\} \cup \{b_j : j = 1, 2, \dots, d_1 + d_3 + d_5 + \dots\}$,*
- (iii) *the edges of G' have labels $1, 2, 3, \dots, t - 1$.*

Proof. Let $G = S(d_1, d_2, \dots, d_v)$ be a caterpillar of order t . Then $t = 1 + d_1 + d_2 + \dots + d_v$. Let G' be the subgraph of $C_{n,t-1}$ induced by the following edge set

$$\begin{aligned} & \{a_0 b_i : i = 1, 2, \dots, d_1\} \\ \cup & \{a_i b_{d_1} : i = -1, -2, \dots, -d_2\} \\ \cup & \{a_{-d_2} b_i : i = d_1 + 1, d_1 + 2, \dots, d_1 + d_3\} \\ \cup & \{a_i b_{d_1+d_3} : i = -d_2 - 1, -d_2 - 2, \dots, -d_2 - d_4\} \\ \cup & \{a_{-d_2-d_4} b_i : i = d_1 + d_3 + 1, d_1 + d_3 + 2, \dots, d_1 + d_3 + d_5\} \\ \cup & \dots \end{aligned}$$

We see that G' has the required properties. $\square$

For a subgraph S of $C_{n,k}$ and an integer v , let $S+v$ denote the subgraph of $C_{n,k}$ which is induced by $\{a_{i+v} b_{j+v} : a_i b_j \in E(S)\}$.

Lemma 2.2 *Let G be a balanced caterpillar of order t . Suppose that n is a positive integer $\geq t$ such that $n \equiv 0 \pmod{t/2}$. Then $C_{n,t-1}$ has a G -factorization.*

Proof. Let $G = S(d_1, d_2, \dots, d_v)$. Since G is balanced and of order t , $d_1 + d_3 + d_5 + \dots = 1 + d_2 + d_4 + d_6 + \dots = t/2$. Let $n' = \frac{n}{t/2}$. Let G' be the subgraph of $C_{n,t-1}$ described in Lemma 2.1. Let F be the subgraph of $C_{n,t-1}$ such that

$$F = G' \cup (G' + \frac{t}{2}) \cup (G' + 2 \cdot \frac{t}{2}) \cup \dots \cup (G' + (n' - 1) \frac{t}{2}).$$

From Lemma 2.1(ii), we see that F is a G -factor of $C_{n,t-1}$. From Lemma 2.1(iii), we see that $C_{n,t-1}$ can be decomposed into $F, F+1, F+2, \dots, F+(\frac{t}{2}-1)$. Hence $C_{n,t-1}$ has a G -factorization. $\square$

Now we have the necessary and sufficient conditions for the balanced caterpillar factorization of crowns.

Theorem 2.3 *Let G be a balanced caterpillar of order t . Suppose that $n \geq k$ are positive integers. Then $C_{n,k}$ has a G -factorization if and only if $n \equiv 0 \pmod{t/2}$ and $k \equiv 0 \pmod{t-1}$.*

Proof. (*Necessity*) Note that $C_{n,k}$ has $2n$ vertices, nk edges, and G has t vertices, $t-1$ edges where t is an even integer since G is balanced. By Lemma 1.1 (1) and (2), we have

$$2n \equiv 0 \pmod{t}, \quad t \cdot nk \equiv 0 \pmod{2n(t-1)}.$$

Thus $n \equiv 0 \pmod{t/2}$, and $tk \equiv 0 \pmod{t-1}$. The latter implies $k \equiv 0 \pmod{t-1}$ since t and $t-1$ are relatively prime.

(*Sufficiency*) Since $n \equiv 0 \pmod{t/2}$, it follows from Lemma 2.2 that $C_{n,t-1}$ has a G -factorization. Since $k \equiv 0 \pmod{t-1}$, it follows from Lemma 1.3 that $C_{n,k}$ has a G -factorization. $\square$

The problem for nonbalanced-caterpillar factorization of crowns remains open.

3 Directed caterpillar factorization of symmetric crowns

In this section the directed caterpillar factorization of symmetric crowns is considered. We assign the arc $\overrightarrow{a_i b_j}$ and the arc $\overrightarrow{b_j a_i}$ in $C_{n,k}^*$ with the same label of $a_i b_j$ in $C_{n,k}$, i.e., for $1 \leq t \leq k$, $\overrightarrow{a_i b_{i+t}}$ has label t , and $\overrightarrow{b_{i+t} a_i}$ also has label t .

Suppose that G is a subdigraph of $C_{n,k}^*$ and G has no isolated vertex. The *dual* of G , denoted by $Dual(G)$, is the subdigraph of $C_{n,k}^*$ induced by the arc set $\{\overrightarrow{b_{-i} a_{-j}} : \overrightarrow{a_i b_j} \in E(G)\} \cup \{\overrightarrow{a_{-j} b_{-i}} : \overrightarrow{b_j a_i} \in E(G)\}$. Note that $Dual(G) \cong G$. Note also that the arc $\overrightarrow{a_i b_j}$ in G , which is from A to B , and the arc $\overrightarrow{b_{-i} a_{-j}}$ in $Dual(G)$, which is from B to A , have the same label. Similarly, the arc $\overrightarrow{b_j a_i}$ in G , which is from B to A , and the arc $\overrightarrow{a_{-j} b_{-i}}$ in $Dual(G)$, which is from A to B , have the same label.

For a subdigraph S of $C_{n,k}^*$ and an integer v , let $S+v$ denote the subdigraph of $C_{n,k}^*$ which is induced by $\{\overrightarrow{a_{i+v} b_{j+v}} : \overrightarrow{a_i b_j} \in E(S)\}$.

Lemma 3.1 *Let G be a caterpillar of order t . Suppose that $\hat{G}$ is a digraph obtained from G by orienting each edge of G arbitrarily. We have the following*

- (1) *If G is balanced, and n is an integer $\geq t$ such that $n \equiv 0 \pmod{t/2}$, then $C_{n,t-1}^*$ has a $\hat{G}$ -factorization.*
- (2) *If G is unbalanced, and n is a positive integer such that $n \equiv 0 \pmod{t}$, then $C_{n,t-1}^*$ has a $\hat{G}$ -factorization.*

Proof. Let $G = S(d_1, d_2, \dots, d_v)$ be a caterpillar of order t . By Lemma 2.1, there exists a subgraph G' of the crown $C_{n,t-1}$ such that

- (i) $G' \cong G$,
- (ii) the vertex set of $G' = \{a_i : i = 0, -1, -2, \dots, -d_2 - d_4 - d_6 - \dots\} \cup \{b_j : j = 1, 2, \dots, d_1 + d_3 + d_5 + \dots\}$,
- (iii) the edges of G' have labels $1, 2, 3, \dots, t-1$.

Since $G' \cong G$, we can orient the edges of G' so that the resulting digraph, say G'' , is isomorphic to $\hat{G}$. Hence G'' is a subdigraph of $C_{n,t-1}^*$ such that

- (i) $G'' \cong \hat{G}$,
- (ii) the vertex set of $G'' = \{a_i : i = 0, -1, -2, \dots, -d_2 - d_4 - d_6 - \dots\} \cup \{b_j : j = 1, 2, \dots, d_1 + d_3 + d_5 + \dots\}$,
- (iii) the arcs of G'' have labels $1, 2, 3, \dots, t-1$.

Let $G''' = Dual(G'')$. Then

- (i) $G''' \cong \hat{G}$,
- (ii) the vertex set of $G''' = \{a_i : i = -1, -2, \dots, -d_1 - d_3 - d_5 - \dots\} \cup \{b_j : j = 0, 1, 2, \dots, d_2 + d_4 + d_6 + \dots\}$,
- (iii) the arcs of G''' have labels $1, 2, 3, \dots, t-1$.

Now we prove (1). Since G is balanced, we have $d_1 + d_3 + d_5 + \dots = 1 + d_2 + d_4 + \dots = t/2$. Let $n' = \frac{n}{t/2}$. Let $F_1 = G''' \cup (G''' + \frac{t}{2}) \cup (G''' + 2 \cdot \frac{t}{2}) \cup (G''' + 3 \cdot \frac{t}{2}) \cup \dots \cup (G''' + (n'-1) \cdot \frac{t}{2})$. Then F_1 is a $\hat{G}$ -factor of $C_{n,t-1}^*$. Let $F_2 = G''' \cup (G''' + \frac{t}{2}) \cup (G''' + 2 \cdot \frac{t}{2}) \cup (G''' + 3 \cdot \frac{t}{2}) \cup \dots \cup (G''' + (n'-1) \cdot \frac{t}{2})$. Then F_2 is a $\hat{G}$ -factor of $C_{n,t-1}^*$. We can see that $C_{n,t-1}^*$ can be decomposed into the following $\hat{G}$ -factors: $F_1, F_1+1, F_1+2, \dots, F_1+(\frac{t}{2}-1)$ and $F_2, F_2+1, F_2+2, \dots, F_2+(\frac{t}{2}-1)$.

Next we prove (2). Let $G'''' = G''' + (1 + d_1 + d_3 + d_5 + \dots)$. Then

- (i) $G'''' \cong \hat{G}$,
- (ii) the vertex set of G'''' is $\{a_i : i = 1, 2, \dots, -1 + d_1 + d_3 + d_5 + \dots, d_1 + d_3 + d_5 + \dots\} \cup \{b_j : j = 1 + d_1 + d_3 + d_5 + \dots, 2 + d_1 + d_3 + d_5 + \dots, \dots, 1 + d_1 + d_2 + d_3 + d_4 + \dots\}$,
- (iii) the arcs of G'''' have labels $1, 2, 3, \dots, t-1$.

Let $n' = \frac{n}{t}$. Let $F = G'' \cup (G'' + t) \cup (G'' + 2t) \cup \dots \cup (G'' + (n'-1)t) \cup G'''' \cup (G'''' + t) \cup (G'''' + 2t) \cup \dots \cup (G'''' + (n'-1)t)$. Then F is a $\hat{G}$ -factor of $C_{n,t-1}^*$, and $C_{n,k}^*$ is decomposed into the following $\hat{G}$ -factors: $F, F+1, F+2, \dots, F+(t-1)$. $\square$

Now we have the necessary and sufficient conditions for the directed caterpillar factorization of symmetric crowns.

Theorem 3.2 *Let G be a caterpillar of order t , and $\hat{G}$ be a digraph obtained from G by orienting each edge of G arbitrarily. Let $n \geq k$ be positive integers. We have the following*

- (1) *Suppose that G is balanced. Then $C_{n,k}^*$ has a $\hat{G}$ -factorization if and only if $n \equiv 0 \pmod{t/2}$, $k \equiv 0 \pmod{t-1}$.*
- (2) *Suppose that G is unbalanced. Then $C_{n,k}^*$ has a $\hat{G}$ -factorization if and only if $n \equiv 0 \pmod{t}$, $k \equiv 0 \pmod{t-1}$.*

Proof. We first prove (1).

(Necessity) Note that $C_{n,k}^*$ has $2n$ vertices, $2nk$ arcs, and $\hat{G}$ has t vertices, $t-1$ arcs where t is an even integer since G is balanced. By Lemma 1.1 (1) and (2), $2n \equiv 0 \pmod{t}$, $t \cdot 2nk \equiv 0 \pmod{2n(t-1)}$. Hence $n \equiv 0 \pmod{t/2}$, $tk \equiv 0 \pmod{t-1}$. The latter implies $k \equiv 0 \pmod{t-1}$.

(Sufficiency) From the assumption, we have $n \equiv 0 \pmod{t/2}$, $n \geq t$ (for $n \geq k \geq 0$, $k \equiv 0 \pmod{t-1}$, $n \equiv 0 \pmod{t/2}$). By Lemma 3.1(1), $C_{n,t-1}^*$ has a $\hat{G}$ -factorization. Since $n \geq k$ and $k \equiv 0 \pmod{t-1}$, $C_{n,k}^*$ has a $\hat{G}$ -factorization by Lemma 1.3.

Next we prove (2).

(Necessity) As in the proof of the necessity of (1), by Lemma 1.1(2) we have $t \cdot 2nk \equiv 0 \pmod{2n(t-1)}$, and hence $k \equiv 0 \pmod{t-1}$. The existence of $\hat{G}$ -factorization of $C_{n,k}^*$ implies that of $\hat{G}$ -factor of $C_{n,k}^*$, and hence implies that of G -factor of $C_{n,k}$. Since $C_{n,k}$ is a balanced bipartite graph of order $2n$, and G is an unbalanced

bipartite subgraph of $C_{n,k}$ with order t , by Lemma 1.2 we have $n \equiv 0 \pmod{t}$.

(Sufficiency) Since G is an unbalanced caterpillar of order t and n is a positive integer with $n \equiv 0 \pmod{t}$, by Lemma 3.1(2) $C_{n,t-1}^*$ has a $\hat{G}$ -factorization. Since $k \equiv 0 \pmod{t-1}$ and $k \leq n$, $C_{n,k}^*$ has a $\hat{G}$ -factorization by Lemma 1.3. $\square$

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